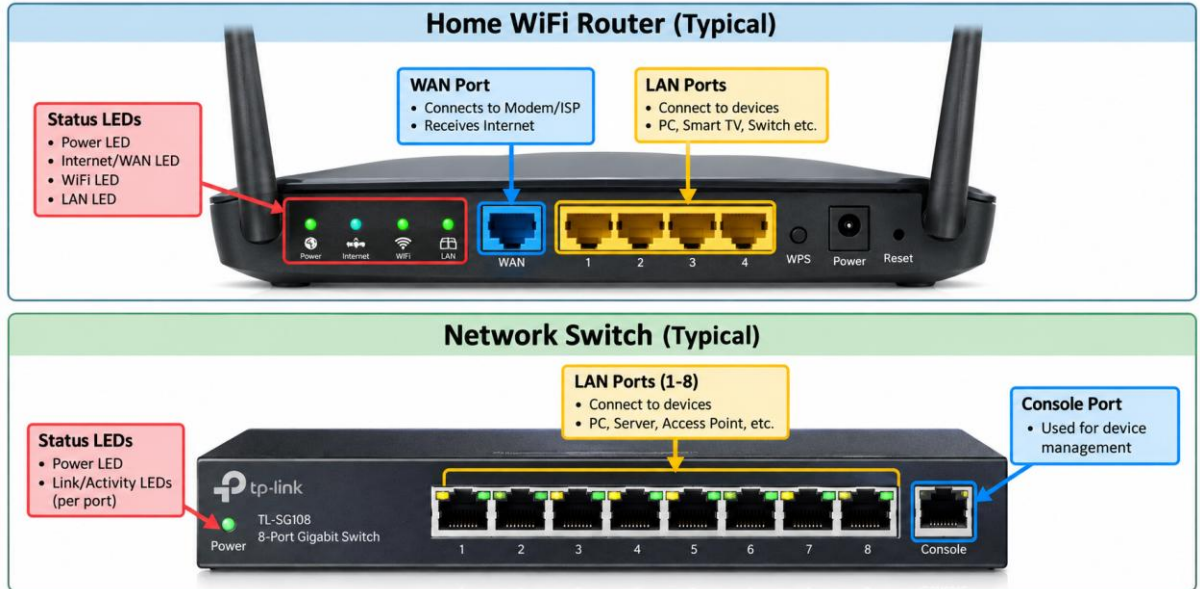


Routers and Switches Theory

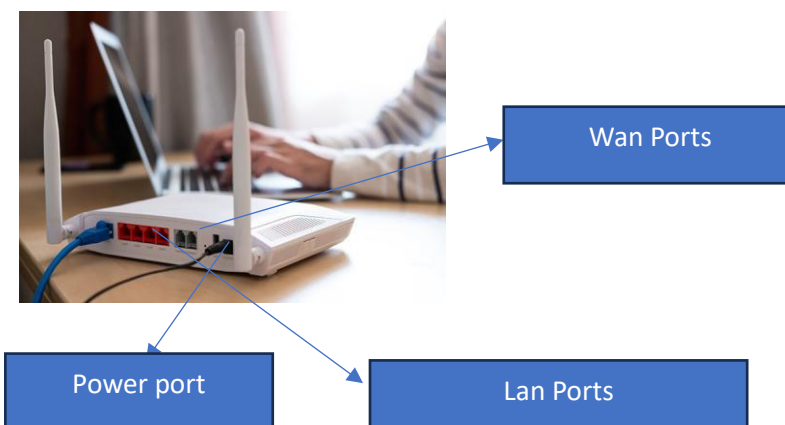
1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Router	Switch
Connects different networks and forwards packets between them.	Connects devices within the same network and forwards frames between them.
Primarily uses IP addresses to determine where traffic should go.	Primarily uses MAC addresses to forward traffic to the correct device.
Routes traffic between different networks, such as a LAN and the Internet/WAN .	Handles traffic within a LAN , such as communication between PCs, printers, and servers.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Router Ports

Power Port: The power port is used to connect the router to its power adapter and provide electricity to the device.

WAN Port: The WAN port connects the router to the modem or Internet service provider, allowing the router to access the Internet.

LAN Port: LAN ports are used to connect wired devices such as computers, printers, or a network switch to the local network.



Switch Ports

Power Port: The power port is located on the back side of the D-Link switch and is used to connect the switch to a power adapter, providing power to the device.

Ethernet Ports: This switch has **8 Ethernet ports**, numbered 1 to 8 on the front panel. These ports are used to connect computers, printers, routers, servers, and other network devices within the local network.

LED Indicators: The LED indicators are located above the Ethernet ports. They show the **power status and link/activity status** of each connected Ethernet port, indicating whether a device is connected and transmitting data.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Small Gaming Café Network Scenario

For a small gaming café, I would connect the **ISP line first to the router's WAN port**. The router receives the Internet connection from the ISP and manages the network using IP addresses.

Then, I would connect one **LAN port of the router to the network switch**. The switch would then be used to connect the **8 gaming PCs** using eight Ethernet cables, with one PC connected to each switch port.

Connection:

ISP → Router (WAN) → Router (LAN) → Switch → 8 Gaming PCs

Justification: The **router** provides Internet access and manages communication between the café's local network and the Internet, while the **switch** provides multiple Ethernet ports so all 8 PCs can communicate efficiently on the local

network. This setup is suitable for IPL streaming and multiplayer gaming because the PCs get a shared Internet connection through the router.